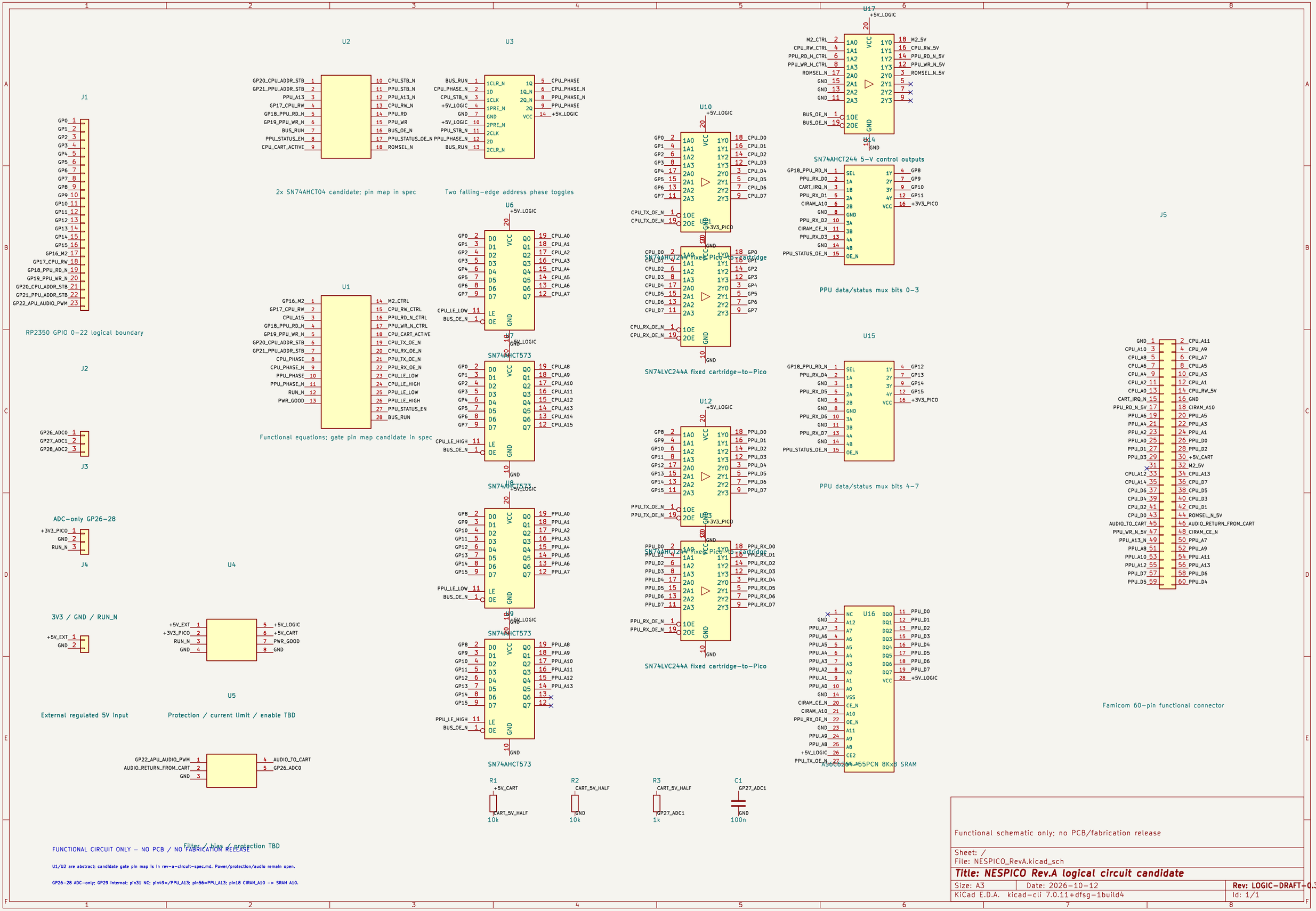

J1
GP0 1
GP1 2
GP2 3
GP3 4
GP4 5
GP5 6
GP6 7
GP7 8
GP8 9
GP9 10
GP10 11
GP11 12
GP12 13
GP13 14
GP14 15
GP15 16
GP16_M2 17
GP17_CPU_RW 18
GP18_PPU_RD_N 19
GP19_PPU_WR_N 20
GP20_CPU_ADDR_STB 21
GP21_PPU_ADDR_STB 22
GP22_APU_AUDIO_PWM 23
RP2350 GPIO 0–22 logical boundary
J2
GP26_ADC0 1
GP27_ADC1 2
GP28_ADC2 3
J3
ADC-only GP26-28
+3V3_PICO 1
GND 2
RUN_N 3
J4
3V3 / GND / RUN_N
+5V_EXT 1
GND 2
External regulated 5V input
U2
GP20_CPU_ADDR_STB 1
GP21_PPU_ADDR_STB 2
PPU_A13 3
GP17_CPU_RW 4
GP18_PPU_RD_N 5
GP19_PPU_WR_N 6
BUS_RUN 7
PPU_STATUS_EN 8
CPU_CART_ACTIVE 9
10 CPU_STB_N
11 PPU_STB_N
12 PPU_A13_N
13 CPU_RW_N
14 PPU_RD
15 PPU_WR
16 BUS_OE_N
17 PPU_STATUS_OE_N
18 ROMSEL_N
2x SN74AHCT04 candidate; pin map in spec
U3
BUS_RUN 1
CPU_PHASE_N 2
CPU_STB_N 3
+5V_LOGIC 4
GND 7
+5V_LOGIC 10
PPU_STB_N 11
PPU_PHASE_N 12
BUS_RUN 13
1CLR_N
1D
1CLK
1PRE_N
GND
2PRE_N
2CLK
2D
2CLR_N
1Q
1Q_N
2Q_N
2Q
VCC
5 CPU_PHASE
6 CPU_PHASE_N
8 PPU_PHASE_N
9 PPU_PHASE
14 +5V_LOGIC
Two falling-edge address phase toggles
U1
GP16_M2 1
GP17_CPU_RW 2
CPU_A15 3
GP18_PPU_RD_N 4
GP19_PPU_WR_N 5
GP20_CPU_ADDR_STB 6
GP21_PPU_ADDR_STB 7
CPU_PHASE 8
CPU_PHASE_N 9
PPU_PHASE 10
PPU_PHASE_N 11
RUN_N 12
PWR_GOOD 13
14 M2_CTRL
15 CPU_RW_CTRL
16 PPU_RD_N_CTRL
17 PPU_WR_N_CTRL
18 CPU_CART_ACTIVE
19 CPU_TX_OE_N
20 CPU_RX_OE_N
21 PPU_TX_OE_N
22 PPU_RX_OE_N
23 CPU_LE_LOW
24 CPU_LE_HIGH
25 PPU_LE_LOW
26 PPU_LE_HIGH
27 PPU_STATUS_EN
28 BUS_RUN
Functional equations; gate pin map candidate in spec
U6
+5V_LOGIC
GP0 2
GP1 3
GP2 4
GP3 5
GP4 6
GP5 7
GP6 8
GP7 9
CPU_LE_LOW 11
BUS_OE_N 1
D0 D1 D2 D3 D4 D5 D6 D7
LE
OE
VCC
GND
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
19 CPU_A0
18 CPU_A1
17 CPU_A2
16 CPU_A3
15 CPU_A4
14 CPU_A5
13 CPU_A6
12 CPU_A7
SN74AHCT573
U7
+5V_LOGIC
GP8 2
GP9 3
GP10 4
GP11 5
GP12 6
GP13 7
GP14 8
GP15 9
CPU_LE_HIGH 11
BUS_OE_N 1
19 CPU_A8
18 CPU_A9
17 CPU_A10
16 CPU_A11
15 CPU_A12
14 CPU_A13
13 CPU_A14
12 CPU_A15
SN74AHCT573
U8
+5V_LOGIC
GP8 2
GP9 3
GP10 4
GP11 5
GP12 6
GP13 7
GP14 8
GP15 9
PPU_LE_LOW 11
BUS_OE_N 1
19 PPU_A8
18 PPU_A9
17 PPU_A10
16 PPU_A11
15 PPU_A12
14 PPU_A13
SN74AHCT573
U4
U5
+5V_EXT 1
+3V3_PICO 2
RUN_N 3
GND 4
5 +5V_LOGIC
6 +5V_CART
7 PWR_GOOD
8 GND
Protection / current limit / enable TBD
GP22_APU_AUDIO_PWM 1
AUDIO_RETURN_FROM_CART 2
GND 3
4 AUDIO_TO_CART
5 GP26_ADC0
Filter / bias / protection TBD
U10
+5V_LOGIC
GP0 2
GP1 4
GP2 6
GP3 8
GP4 17
GP5 15
GP6 13
GP7 11
CPU_TX_OE_N 1
CPU_TX_OE_N 19
1A0 1A1 1A2 1A3 2A0 2A1 2A2 2A3
1OE
2OE
VCC
GND
1Y0 1Y1 1Y2 1Y3 2Y0 2Y1 2Y2 2Y3
18 CPU_D0
16 CPU_D1
14 CPU_D2
12 CPU_D3
3 CPU_D4
5 CPU_D5
7 CPU_D6
9 CPU_D7
+3V3_PICO
SN74AHCT244 fixed Pico-to-cartridge
CPU_D0 2
CPU_D1 4
CPU_D2 6
CPU_D3 8
CPU_D4 17
CPU_D5 15
CPU_D6 13
CPU_D7 11
CPU_RX_OE_N 1
CPU_RX_OE_N 19
18 GP0
16 GP1
14 GP2
12 GP3
3 GP4
5 GP5
7 GP6
9 GP7
SN74LVC244A fixed cartridge-to-Pico
U12
+5V_LOGIC
GP8 2
GP9 4
GP10 6
GP11 8
GP12 17
GP13 15
GP14 13
GP15 11
PPU_TX_OE_N 1
PPU_TX_OE_N 19
18 PPU_D0
16 PPU_D1
14 PPU_D2
12 PPU_D3
3 PPU_D4
5 PPU_D5
7 PPU_D6
9 PPU_D7
+3V3_PICO
SN74AHCT244 fixed Pico-to-cartridge
PPU_D0 2
PPU_D1 4
PPU_D2 6
PPU_D3 8
PPU_D4 17
PPU_D5 15
PPU_D6 13
PPU_D7 11
PPU_RX_OE_N 1
PPU_RX_OE_N 19
18 PPU_RX_D0
16 PPU_RX_D1
14 PPU_RX_D2
12 PPU_RX_D3
3 PPU_RX_D4
5 PPU_RX_D5
7 PPU_RX_D6
9 PPU_RX_D7
SN74LVC244A fixed cartridge-to-Pico
U17
+5V_LOGIC
M2_CTRL 2
CPU_RW_CTRL 4
PPU_RD_N_CTRL 6
PPU_WR_N_CTRL 8
ROMSEL_N 17
GND 15
GND 13
GND 11
BUS_OE_N 1
BUS_OE_N 19
1A0 1A1 1A2 1A3 2A0 2A1 2A2 2A3
1OE
2OE
VCC
GND
1Y0 1Y1 1Y2 1Y3 2Y0 2Y1 2Y2 2Y3
18 M2_5V
16 CPU_RW_5V
14 PPU_RD_N_5V
12 PPU_WR_N_5V
3 ROMSEL_N_5V
5
7
9
SN74AHCT244 5-V control outputs
GP18_PPU_RD_N 1
PPU_RX_D0 2
CART_IRQ_N 3
PPU_RX_D1 5
CIRAM_A10 6
GND 8
PPU_RX_D2 10
CIRAM_CE_N 11
PPU_RX_D3 13
GND 14
PPU_STATUS_OE_N 15
SEL
1A 1B 2A 2B 3A 3B 4A 4B
OE_N
VCC
GND
1Y 2Y 3Y 4Y
4 GP8
7 GP9
9 GP10
12 GP11
16 +3V3_PICO
PPU data/status mux bits 0–3
U15
GP18_PPU_RD_N 1
PPU_RX_D4 2
GND 3
PPU_RX_D5 5
GND 6
GND 8
PPU_RX_D6 10
GND 11
PPU_RX_D7 13
GND 14
PPU_STATUS_OE_N 15
4 GP12
7 GP13
9 GP14
12 GP15
16 +3V3_PICO
PPU data/status mux bits 4–7
U16
AS6C6264-55PCN 8Kx8 SRAM
NC GND A7 A6 A5 A4 A3 A2 A1 A0 VSS CE_N A10 OE_N A11 A9 A8 CE2 VCC
DQ0 DQ1 DQ2 DQ3 DQ4 DQ5 DQ6 DQ7
11 PPU_D0
12 PPU_D1
13 PPU_D2
15 PPU_D3
16 PPU_D4
17 PPU_D5
18 PPU_D6
19 PPU_D7
28 +5V_LOGIC
J5
GND 1
CPU_A10 3
CPU_A8 5
CPU_A6 7
CPU_A4 9
CPU_A2 11
CPU_A0 13
CART_IRQ_N 15
PPU_RD_N_5V 17
PPU_A6 19
PPU_A4 21
PPU_A2 23
PPU_A0 25
PPU_D1 27
PPU_D3 29
31
CPU_A12 33
CPU_A14 35
CPU_D6 37
CPU_D4 39
CPU_D2 41
CPU_D0 43
AUDIO_TO_CART 45
PPU_WR_N_5V 47
PPU_A13_N 49
PPU_A8 51
PPU_A10 53
PPU_A12 55
PPU_D7 57
PPU_D5 59
2 CPU_A11
4 CPU_A9
6 CPU_A7
8 CPU_A5
10 CPU_A3
12 CPU_A1
14 CPU_RW_5V
16 GND
18 CIRAM_A10
20 PPU_A5
22 PPU_A3
24 PPU_A1
26 PPU_D0
28 PPU_D2
30 +5V_CART
32 M2_5V
34 CPU_A13
36 CPU_D7
38 CPU_D5
40 CPU_D3
42 CPU_D1
44 ROMSEL_N_5V
46 AUDIO_RETURN_FROM_CART
48 CIRAM_CE_N
50 PPU_A7
52 PPU_A9
54 PPU_A11
56 PPU_A13
58 PPU_D6
60 PPU_D4
Famicom 60-pin functional connector
R1
+5V_CART
CART_5V_HALF
10k
R2
CART_5V_HALF
GND
10k
R3
CART_5V_HALF
GP27_ADC1
1k
C1
GP27_ADC1
GND
100n
FUNCTIONAL CIRCUIT ONLY — NO PCB / NO FABRICATION RELEASE
U1/U2 are abstract; candidate gate pin map is in rev-a-circuit-spec.md. Power/protection/audio remain open.
GP26-28 ADC-only; GP29 internal; pin31 NC; pin49=/PPU_A13; pin56=PPU_A13; pin18 CIRAM_A10 -> SRAM A10.
Functional schematic only; no PCB/fabrication release
Sheet: /
File: NESPICO_RevA.kicad_sch
Title: NESPICO Rev.A logical circuit candidate
Size: A3
Date: 2026-10-12
Rev: LOGIC-DRAFT-0.3
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
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